

MATB34 - Written Assignment

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1 Laplace transforms

(1)

$$\mathcal{L}\{\cos^2(\omega t)\} = \mathcal{L}\left\{\frac{1}{2}(1 + \cos(2\omega t))\right\} = \frac{1}{2}(\mathcal{L}\{1\} + \mathcal{L}\{\cos(2\omega t)\}) = \frac{1}{2}\left(\frac{1}{s} + \frac{s}{s^2 + 4\omega^2}\right).$$

(2)

$$\mathcal{L}\{t^2 - 1\} = \mathcal{L}\{t^2\} - \mathcal{L}\{1\} = \frac{2}{s^3} - \frac{1}{s}.$$

(3)

$$\mathcal{L}\{e^{-t} \cos^2 t\} = \mathcal{L}\left\{e^{-t} \cdot \frac{1}{2}(1 + \cos(2t))\right\} = \frac{1}{2} \mathcal{L}\{e^{-t}\} + \frac{1}{2} \mathcal{L}\{e^{-t} \cos(2t)\} = \frac{1}{2(s+1)} + \frac{s+1}{2((s+1)^2 + 4)}.$$

2 Inverse Laplace transforms

(1)

$$\begin{aligned} \mathcal{L}^{-1}\left\{\frac{1}{(s-a)(s-b)}\right\} &= \mathcal{L}^{-1}\left\{\frac{1}{a-b}\left(\frac{1}{s-a} - \frac{1}{s-b}\right)\right\} \\ &= \frac{1}{a-b} \mathcal{L}^{-1}\left\{\frac{1}{s-a} - \frac{1}{s-b}\right\} \\ &= \frac{1}{a-b} \left(\mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s-b}\right\}\right) \\ &= \frac{1}{a-b} (e^{at} - e^{bt}) \\ &= \frac{e^{at} - e^{bt}}{a-b} \quad (a \neq b). \end{aligned}$$

$$\text{If } a = b \text{ then } \mathcal{L}^{-1}\left\{\frac{1}{(s-a)^2}\right\} = te^{at}$$

(2)

$$\begin{aligned} \mathcal{L}^{-1}\left\{\frac{s^2}{1+s^2}\right\} &= \mathcal{L}^{-1}\left\{\frac{s^2+1-1}{s^2+1}\right\} \\ &= \mathcal{L}^{-1}\left\{1 - \frac{1}{s^2+1}\right\} \\ &= \mathcal{L}^{-1}\{1\} - \mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} \\ &= \delta(t) - \sin t. \end{aligned}$$

3 Ordinary differential equations

(1)

$$y'' - y' + 2y = 2e^{3t}, \quad y(0) = 0, \quad y'(0) = 0.$$

$$(s^2Y - sy(0) - y'(0)) - (sY - y(0)) + 2Y = \frac{2}{s-3}$$

$$(s^2 - s + 2)Y = \frac{2}{s-3} \Rightarrow Y = \frac{2}{(s-3)(s^2 - s + 2)}.$$

$$Y = \frac{1}{4} \frac{1}{s-3} - \frac{1}{4} \frac{s+2}{s^2 - s + 2} = \frac{1}{4} \frac{1}{s-3} - \frac{1}{4} \frac{s - \frac{1}{2}}{(s - \frac{1}{2})^2 + \frac{7}{4}} - \frac{5}{4} \frac{1}{(s - \frac{1}{2})^2 + \frac{7}{4}}.$$

$$\Rightarrow y(t) = \frac{1}{4}e^{3t} - \frac{1}{4}e^{t/2} \cos \frac{\sqrt{7}t}{2} - \frac{5}{4\sqrt{7}}e^{t/2} \sin \frac{\sqrt{7}t}{2}.$$

(2)

$$y'' + 4y' + 4y = t^3 * e^{-2t}, \quad y(0) = 1, \quad y'(0) = 2.$$

$$(s^2Y - sy(0) - y'(0)) + 4(sY - y(0)) + 4Y = \frac{6}{s^4} \cdot \frac{1}{s+2}$$

$$(s+2)^2Y = (s+6) + \frac{6}{s^4(s+2)} \Rightarrow Y = \frac{s+6}{(s+2)^2} + \frac{6}{s^4(s+2)^3}.$$

$$\frac{s+6}{(s+2)^2} = \frac{1}{s+2} + \frac{4}{(s+2)^2},$$

$$\frac{6}{s^4(s+2)^3} = \frac{-\frac{15}{16}}{s} + \frac{\frac{9}{8}}{s^2} - \frac{\frac{9}{8}}{s^3} + \frac{\frac{3}{4}}{s^4} + \frac{\frac{15}{16}}{s+2} + \frac{\frac{3}{4}}{(s+2)^2} + \frac{\frac{3}{8}}{(s+2)^3}.$$

$$\Rightarrow y(t) = \frac{31}{16}e^{-2t} + \frac{19}{4}te^{-2t} + \frac{3}{16}t^2e^{-2t} - \frac{15}{16} + \frac{9}{8}t - \frac{9}{16}t^2 + \frac{1}{8}t^3$$